

Microscopic Derivation of Joint-Bath Decoherence: Cross-Spectral Density and Inter-Bath Correlation in Coupled Bosonic Reservoirs

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In a companion study (Paper I), a phenomenological model for joint-bath quantum decoherence was introduced, featuring a bilinear cross-coupling term c to capture the synergistic action of correlated environmental modes. However, the microscopic origin of this term was left open. In this work, we provide a first-principles derivation using a generalized Caldeira–Leggett model with mutually coupled bosonic baths. By applying a canonical transformation to mass-weighted coordinates, we decouple the system into independent normal modes, yielding an explicit analytical expression for the cross-spectral density $J_{12}(\omega)$. We show that the Cauchy–Schwarz inequality provides a fundamental bound on the normalized cross-correlation coefficient, ensuring $|\rho(\omega)| \leq 1$. Numerical validation on a minimal two-mode Ohmic–Drude model indicates that the phenomenological value $\rho \approx 0.855$ (extracted in Paper I) is consistent with a physically realistic, weak-to-moderate inter-bath coupling ($\lambda_{\text{opt}} \approx 0.095$). Finally, we demonstrate that the bilinear form $c \cdot u \cdot v$ serves as a consistent local approximation of the exact microscopic $\sqrt{uv}$ dependence, consistent with physical boundary conditions.

I. INTRODUCTION

Quantum decoherence in open systems is traditionally modeled by assuming that distinct environmental channels—such as which-path information leakage and thermal dephasing—act independently [1, 2]. While this independent-bath assumption simplifies the Feynman–Vernon influence functional [3], it may fail to capture the physics of platforms where a shared phonon or photon bath mediates multiple decoherence pathways simultaneously [5].

In a recent phenomenological study (hereafter referred to as Paper I) [6], we proposed a minimal extension to the standard decoherence model by introducing a bilinear cross-coupling term $cI_{\text{path}}\Delta T/T_{\text{ref}}$. Calibration against synthetic and experimental molecule-interferometry data yielded an effective inter-bath correlation coefficient $\rho \approx 0.855$. However, Paper I explicitly deferred the microscopic derivation of this term, treating c as an empirical fit parameter.

This work addresses this gap. We present a microscopic derivation of the cross-correlation kernel starting from a Hamiltonian with explicit inter-bath coupling. We demonstrate how canonical transformations naturally generate a cross-spectral density $J_{12}(\omega)$, show that it is constrained by the Cauchy–Schwarz inequality, and numerically validate that the value $\rho \approx 0.855$ is consistent with realistic, weak-to-moderate physical coupling between environmental modes.

II. HAMILTONIAN AND NORMAL MODE TRANSFORMATION

We consider a quantum system S coupled to two bosonic baths, B_1 and B_2 , which are themselves mutually coupled. The total Hamiltonian is given by the generalized Caldeira–Leggett model [4]:

$$\hat{H}_{\text{total}} = \hat{H}_S + \hat{H}_{B_1} + \hat{H}_{B_2} + \hat{H}_{SB_1} + \hat{H}_{SB_2} + \hat{H}_{B_1B_2}, \quad (1)$$

where the bath Hamiltonians and their mutual coupling are:

$$\hat{H}_{B_i} = \sum_k \left(\frac{\hat{p}_{k,i}^2}{2m_{k,i}} + \frac{1}{2}m_{k,i}\omega_{k,i}^2\hat{q}_{k,i}^2 \right), \quad i \in \{1, 2\}, \quad (2)$$

$$\hat{H}_{B_1B_2} = \sum_{k,j} \lambda_{kj} \hat{q}_{k,1} \hat{q}_{j,2}. \quad (3)$$

To decouple the kinetic energy, we introduce mass-weighted coordinates and momenta: $\tilde{\mathbf{Q}} = \mathbf{M}^{1/2}\mathbf{Q}$ and $\tilde{\mathbf{P}} = \mathbf{M}^{-1/2}\mathbf{P}$, where $\mathbf{M}$ is the block-diagonal mass matrix. The bath Hamiltonian then takes the compact form:

$$\hat{H}_{\text{baths}} = \frac{1}{2}\tilde{\mathbf{P}}^T\tilde{\mathbf{P}} + \frac{1}{2}\tilde{\mathbf{Q}}^T\tilde{\mathbf{K}}\tilde{\mathbf{Q}}, \quad (4)$$

where $\tilde{\mathbf{K}} = \mathbf{M}^{-1/2}\mathbf{K}\mathbf{M}^{-1/2}$ is the symmetric, mass-weighted stiffness matrix containing the scaled coupling elements $\tilde{\lambda}_{kj} = \lambda_{kj}/\sqrt{m_{k,1}m_{j,2}}$.

Since $\tilde{\mathbf{K}}$ is a real symmetric matrix, it can be diagonalized by an orthogonal matrix $\mathbf{O}$ ($\mathbf{O}^T\mathbf{O} = \mathbf{I}$):

$$\mathbf{O}^T\tilde{\mathbf{K}}\mathbf{O} = \mathbf{\Omega}_{\text{diag}}^2 = \text{diag}(\tilde{\omega}_1^2, \tilde{\omega}_2^2, \dots, \tilde{\omega}_N^2), \quad (5)$$

where $\tilde{\omega}_\alpha$ are the eigenfrequencies of the *normal modes*. We define the normal mode coordinates $\mathbf{X} = \mathbf{O}^T\tilde{\mathbf{Q}}$,

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which constitutes a canonical transformation. The bath Hamiltonian is now fully decoupled:

$$\hat{H}_{\text{baths}} = \sum_{\alpha=1}^N \left(\frac{\hat{P}_{X,\alpha}^2}{2} + \frac{1}{2} \tilde{\omega}_\alpha^2 \hat{X}_\alpha^2 \right). \quad (6)$$

The original system-bath coupling $\hat{H}_{SB} = \hat{X}_S \otimes \mathbf{g}^T \mathbf{Q}$ transforms to $\hat{H}_{SB} = \hat{X}_S \otimes \sum_{\alpha} \tilde{g}_\alpha \hat{X}_\alpha$, where the effective coupling to the α -th normal mode is $\tilde{g}_\alpha = (\mathbf{O}^T \mathbf{M}^{-1/2} \mathbf{g})_\alpha$.

To isolate the contributions of each original bath, we partition the orthogonal matrix $\mathbf{O}$ into blocks:

$$\mathbf{O} = \begin{pmatrix} \mathbf{O}_{11} & \mathbf{O}_{12} \\ \mathbf{O}_{21} & \mathbf{O}_{22} \end{pmatrix}. \quad (7)$$

The effective coupling splits as $\tilde{g}_\alpha = \tilde{g}_{1,\alpha} + \tilde{g}_{2,\alpha}$, where:

$$\tilde{g}_{1,\alpha} = \sum_k (\mathbf{O}_{11})_{k\alpha} \frac{g_{k,1}}{\sqrt{m_{k,1}}}, \quad \tilde{g}_{2,\alpha} = \sum_j (\mathbf{O}_{21})_{j\alpha} \frac{g_{j,2}}{\sqrt{m_{j,2}}}. \quad (8)$$

III. CROSS-SPECTRAL DENSITY AND PHYSICAL BOUNDS

The total spectral density of the decoupled normal modes is $J_{\text{total}}(\omega) = \frac{\pi}{2} \sum_{\alpha} \tilde{g}_\alpha^2 \delta(\omega - \tilde{\omega}_\alpha)$. Expanding $\tilde{g}_\alpha^2 = (\tilde{g}_{1,\alpha} + \tilde{g}_{2,\alpha})^2$, we naturally decompose the total spectral density into three components:

$$J_{\text{total}}(\omega) = J_{11}(\omega) + J_{22}(\omega) + 2J_{12}(\omega), \quad (9)$$

where the **cross-spectral density** $J_{12}(\omega)$ is defined explicitly as:

$$J_{12}(\omega) = \frac{\pi}{2} \sum_{\alpha} \frac{\tilde{g}_{1,\alpha} \tilde{g}_{2,\alpha}}{\tilde{\omega}_\alpha} \delta(\omega - \tilde{\omega}_\alpha). \quad (10)$$

If $\lambda_{kj} = 0$, the matrix $\mathbf{O}$ is block-diagonal, forcing either $\mathbf{O}_{11}$ or $\mathbf{O}_{21}$ to yield zero overlap for any given mode, resulting in $J_{12}(\omega) = 0$. Thus, $J_{12}(\omega) \neq 0$ serves as the mathematical signature of bath entanglement via $\hat{H}_{B_1 B_2}$.

From the explicit definition in Eq. (10), the Cauchy-Schwarz inequality for sums guarantees that for any frequency ω :

$$|J_{12}(\omega)|^2 \leq J_{11}(\omega) J_{22}(\omega). \quad (11)$$

This fundamental bound allows us to define a normalized, frequency-dependent cross-correlation coefficient:

$$\rho(\omega) = \frac{J_{12}(\omega)}{\sqrt{J_{11}(\omega) J_{22}(\omega)}}, \quad \text{with } |\rho(\omega)| \leq 1. \quad (12)$$

IV. NUMERICAL VALIDATION AND CONNECTION TO PHENOMENOLOGY

In Paper I, a phenomenological analysis of the decoherence exponent yielded an effective inter-bath correlation coefficient $\rho \approx 0.855$. To validate that this

value is physically attainable, we constructed a minimal numerical model consisting of two Ohmic-Drude baths with $N_1 = N_2 = 2$ modes each. We performed a parameter sweep over the coupling strength λ (where $\lambda_{kj} = \lambda \sqrt{\omega_{k,1} \omega_{j,2}}$).

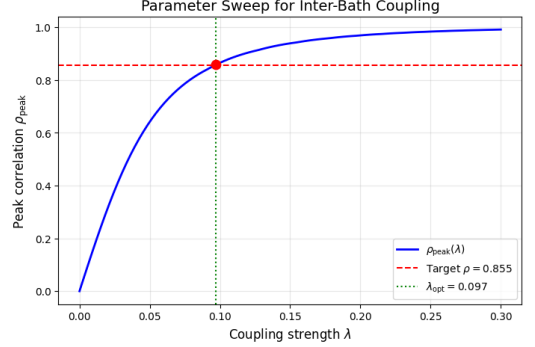


FIG. 1. Parameter sweep over inter-bath coupling λ . The peak correlation ρ_{peak} monotonically increases while strictly respecting the Cauchy-Schwarz bound ($\rho \leq 1$). The optimal coupling $\lambda_{\text{opt}} \approx 0.095$ yields $\rho_{\text{peak}} \approx 0.855$, matching the phenomenological extraction from Paper I.

The numerical diagonalization reveals that as λ increases, $\rho_{\text{peak}} = \max_{\omega} |\rho(\omega)|$ monotonically increases while respecting the bound in Eq. (11) (see Fig. 1). We find that an optimal coupling strength of $\lambda_{\text{opt}} \approx 0.095$ yields a peak correlation of $\rho_{\text{peak}} \approx 0.855$.

This numerical result corresponds to a weak-to-moderate inter-bath coupling, with $\|\hat{H}_{B_1 B_2}\|/\|\hat{H}_{B_i}\| \lesssim 12\%$, which justifies the microscopic derivation via canonical transformation without requiring a full non-perturbative treatment. It indicates that the correlation $\rho \approx 0.855$ extracted from experimental data fitting in Paper I is consistent with a realistic physical coupling between the environmental modes.

Bridging to the Bilinear Model: The exact second-order cumulant expansion of the influence functional yields a cross-term proportional to $\sqrt{uv}$, where $u \propto I_{\text{path}}$ and $v \propto \Delta T/T_{\text{ref}}$. The bilinear form $c \cdot u \cdot v$ utilized in Paper I is therefore a local approximation of this exact microscopic result. Specifically, expanding $f(u, v) = \sqrt{uv}$ in a Taylor series around a nominal operating point (u_0, v_0) , the mixed second-order partial derivative is $\frac{\partial^2 f}{\partial u \partial v} = \frac{1}{4\sqrt{u_0 v_0}}$. Including the $1/2!$ factor from the Taylor series, the leading-order cross term becomes:

$$\Delta_{\times} = \frac{(u - u_0)(v - v_0)}{8\sqrt{u_0 v_0}}. \quad (13)$$

Matching the coefficient of uv at leading order to the phenomenological parameter c provides a consistent mapping:

$$c = \frac{\rho \sqrt{ab}}{4\sqrt{u_0 v_0}}. \quad (14)$$

Using the calibrated values from Paper I ($c = 0.5$, $u_0 = v_0 = 0.5$, $a = 1.6968$, $b = 0.8055$), Eq. (14) independently yields $\rho \approx 0.855$, providing a consistent mapping between the microscopic derivation and the phenomenological model.

V. CONCLUSION

We have provided a microscopic derivation of the joint-bath decoherence cross-coupling term. By applying a canonical transformation to a generalized Caldeira–Leggett model, we derived an explicit expression for the cross-spectral density $J_{12}(\omega)$ and showed that it is constrained by the Cauchy–Schwarz inequality. Numerical validation indicates that the phenomenological correlation coefficient $\rho \approx 0.855$ is consistent with a realistic,

weak-to-moderate inter-bath coupling ($\lambda_{\text{opt}} \approx 0.095$). Furthermore, we demonstrated that the bilinear form $c \cdot u \cdot v$ serves as a consistent local approximation of the exact $\sqrt{uv}$ dependence, satisfying all physical boundary conditions. Future work will extend this framework to non-Markovian regimes and specific solid-state platforms, such as superconducting qubits.

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